Solution Design Document (SDD)

Project: Regulatory Compliance Automation | Platform: Automation Anywhere A360

This document describes a proposed technical design for the daily portal-monitoring bot. Portal-specific selectors, APIs, authentication steps and exact field mappings must be finalized during technical discovery and testing.

1. Solution Overview

An unattended Automation Anywhere A360 bot will run on a configured schedule, load its configuration and UID mapping data, check each approved government portal, capture notice records, validate and deduplicate the data, create a daily report, and write an execution log. The design separates portal-specific retrieval steps from common processing and reporting so that changes can be managed with less rework.

2. Proposed Architecture

Component	Design responsibility
A360 Control Room / scheduler	Schedules the bot, manages deployment and runtime execution, and provides centralized monitoring.
Main orchestration bot	Controls the run lifecycle, loads configuration, calls portal workflows and determines final run status.
Portal adapter subtask(s)	Handles navigation, authentication, search, pagination and extraction for an individual portal.
UID mapping and validation module	Matches retrieved records to approved identifiers and flags missing or ambiguous mappings.
Data store / output folder	Stores the notice register, run log and generated reports in an approved access-controlled location.
Reporting module	Builds daily summary and detail outputs and sends them through the approved delivery channel.
Credential vault	Provides credentials at runtime without hard-coding secrets in the bot or exposing them in logs.

3. Execution Flow

1. Scheduler starts the main bot using the production configuration.
2. Initialize run ID, timestamp, environment and run status.
3. Load portal configuration, search identifiers, UID mapping data and report settings.
4. Validate configuration; if critical data is missing, log a controlled failure and notify support.
5. For each enabled portal, start its portal adapter and establish an authorized session.
6. Run the approved search for each configured UID or entity; handle pagination where required.
7. Extract the notice fields and normalize dates, whitespace, status labels and identifiers.

8. Validate required fields, map UID/entity and calculate a stable duplicate key.
9. Write valid records to the notice register; write incomplete or ambiguous records to the exception register.
10. Aggregate portal-level outcomes, including completed checks, no-notice results and failed checks.
11. Generate the daily summary and detailed report, save outputs and deliver them to the approved recipients.
12. Finalize run status, close browser sessions and write the completion log.

4. Suggested Data Structure

Field	Purpose / handling
Run_ID	Unique identifier for the bot execution.
Portal_Name	Name or code of the portal checked.
Portal_Check_Status	Completed, no results, failed, or partial; must not imply success when a search failed.
UID / Entity_ID	Identifier used for search and mapping.
Notice_Reference	Reference number displayed by the portal, where available.
Notice_Type	Portal label or business-approved classification.
Notice_Date	Date shown by the source, stored in a consistent format.
Due_Date	Response deadline when available; blank if not provided by source.
Notice_Status	Status as displayed by the portal, normalized without changing its meaning.
Source_URL	Link to the relevant notice or search result, if available and permitted.
Retrieved_At	Timestamp of successful retrieval.
Duplicate_Key	Stable key, for example portal + UID/entity + notice reference.
Mapping_Status	Mapped, unmapped or ambiguous.
Exception_Detail	Sanitized description of the issue requiring review.

5. Configuration Design

Keep changeable values in a controlled configuration file or approved configuration store instead of embedding them throughout the workflow.

- Portal code, base URL and enabled/disabled flag.
- Search strategy and portal-specific field/selector configuration.
- Identifier source and mapping-file location.
- Schedule, timeout, retry count and wait intervals.
- Output path, report recipients and file naming convention.
- Retention period, logging level and alert route.

Credentials must be retrieved from the approved credential vault or Control Room credential facility. They must not be stored in plain-text configuration files.

6. Automation Anywhere A360 Build Structure

Bot / task	Responsibilities
BC_Main_Compliance_Monitor	Orchestration, initialization, configuration checks, portal loop, final status and cleanup.
BC_Portal_<PortalCode>	Portal-specific login, search, extraction, pagination and portal-level error handling.
BC_Validate_And_Map	Field validation, date normalization, UID mapping, duplicate checks and exception routing.
BC_Build_Daily_Report	Summary calculations, detail register, exception report and report file creation.
BC_Notify_And_Log	Report delivery, support alerts and final run log updates.

The names above are proposed conventions and can be adjusted to the team's existing naming standard.

7. Error Handling and Recovery

Error condition	Technical response
Timeout / slow page	Use bounded waits and configured retries; if still unsuccessful, mark that portal check as failed.
Authentication failure	Do not loop indefinitely. Log a sanitized error, alert support and skip protected actions.
Element or page structure changed	Use robust selectors where possible; validate that expected page elements are present before extracting data.
No search results	Record a successful check with zero notices only when the search completed and the portal explicitly indicates no results.
Unexpected data format	Validate required fields and send the record to an

	exception output rather than silently discarding it.
Mapping ambiguity	Set Mapping_Status to ambiguous and route the record to the legal/tax review queue.
Duplicate record	Skip duplicate addition to the current report while retaining appropriate audit information.
Report save or email failure	Preserve the report in the approved fallback location and raise an alert.
Bot interruption	Use run ID and processing checkpoints where feasible to support safe restart without duplicating records.

8. Security and Audit

- Use role-based access for Control Room, bot runners, output folders and report recipients.
- Store credentials in an approved vault and retrieve them only at runtime.
- Mask or omit credentials, session tokens and unnecessary personal or confidential data from logs.
- Use approved service accounts and least-privilege access.
- Restrict reports and notice registers to authorized legal, tax, compliance and support users.
- Record run ID, timestamps, portal outcome, record counts, exception counts and report delivery status.
- Apply the organization's retention, backup, encryption and secure deletion policies.

9. Logging and Monitoring

Log event	Minimum information
Run started	Run ID, start timestamp, bot version and environment.
Portal check started/completed	Portal code, timestamp, status and elapsed time.
Records processed	Counts retrieved, valid, duplicate, unmapped and exception records; avoid unnecessary sensitive payloads.
Failure / retry	Portal code, step name, attempt number and sanitized error details.
Report generated	Output file name/path, record counts and generation timestamp.
Notification outcome	Delivery status, recipient group identifier and error details if unsuccessful.
Run completed	End timestamp, overall status and summary counts.

10. Testing Strategy

Test type	Example coverage
Unit / component	Field parsing, date normalization, UID mapping, duplicate key and report calculations.
Portal integration	Successful login, search, pagination, zero-result page and expected notice extraction.
Negative / exception	Invalid credentials, portal timeout, missing fields, ambiguous UID and page layout change.
End-to-end	Scheduled run through retrieval, validation, report generation and delivery.
Recovery	Restart after a partial run and verify that previously processed notices are not duplicated.
Security	Credential masking, role permissions, output-folder access and log review.
User acceptance	Legal/tax stakeholders validate notice accuracy, exception handling and report usefulness.

11. Deployment and Support

13. Complete portal discovery and obtain business, IT and security approvals.
14. Build and test in the development environment using approved test accounts and sample cases.
15. Promote to test/UAT and obtain legal/tax sign-off against agreed acceptance criteria.
16. Deploy the approved package and configuration to production through the established change process.
17. Enable the agreed schedule and verify the first production runs with support monitoring.
18. Document ownership, support hours, alert recipients, retry policy and manual fallback steps.

12. Design Decisions Requiring Confirmation

- Whether each portal supports an approved API, downloadable report or browser-based retrieval.
- Exact authentication requirements and whether multi-factor authentication affects unattended execution.
- Final portal-specific selectors, pagination rules, search limits and rate limits.
- Authoritative mapping source and the duplicate key approved by the business.
- Report format (Excel, CSV or PDF), delivery channel and final report template.
- Production runtime, schedule/time zone, retry limits and monitoring thresholds.
- Data retention, screenshot/diagnostic capture permissions and recovery expectations.

13. Definition of Done

- All approved portals have documented and tested retrieval workflows.
- The bot correctly distinguishes successful zero-result searches from failed checks.
- Notice data, UID mapping, duplicate detection and exception routing pass agreed tests.

- The daily report and audit logs are produced and delivered as approved.
- Security review, UAT sign-off, deployment approval and support handover are complete.